Neelkumar K. Shah

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https://neelkumarkshahportfolio.vercel.app

PROFESSIONAL SUMMARY

Engineer with 5 years of experience in blockchain development, full-stack engineering, and cybersecurity. Built and hardened Hyperledger Fabric and Ethereum systems for India's National Blockchain Service initiative under the Ministry of Electronics and Information Technology (MeitY). Developed a full-stack security assessment platform (React, Node.js, Express.js, MongoDB) and an automated database vulnerability scanner. Authored 5 peer-reviewed IEEE publications on blockchain security.

SKILLS

Blockchain: Hyperledger (Fabric, Sawtooth), Ethereum, Solidity, Truffle, Ganache, IPFS, MetaMask
Development: Python, Node.js, Express.js, NestJS, React, MongoDB, HTML/CSS, Bootstrap
Security: Vulnerability Assessment & Penetration Testing (Web Applications, APIs), Smart Contract Audits (Ethereum, Hyperledger), Static Source Code Analysis, NoSQL Database Security Audit

PROFESSIONAL EXPERIENCE

Self-Employed/Freelance

Independent Full-Stack Developer, Blockchain Engineer & Cybersecurity Consultant | Jun 2026 - Present

- Architected and delivered the full-fledged, production-grade corporate air travel web platform for Travel Beyond tbgind.com - an executive air travel management company with dual operational hubs in Dubai and Mumbai. Developed a high-performance responsive UI, hardened the platform with strict Content Security Policy (CSP), and integrated structured schema.org JSON-LD for global SEO.
- Architected and developed Sudarshana, a full-stack web application security auditing platform built with Node.js, React.js, Express.js, and MongoDB (MERN stack) to automate security assessments for web applications and REST APIs.
- Designed and developed an automated Node.js security scanner that analyzes web applications and APIs for HTTP security headers.
- Developed robust RESTful API endpoints, secure authentication workflows, and interactive React.js dashboards with automated PDF audit report generation, enabling real-time security posture assessment, vulnerability tracking, and remediation management for scanned web applications and API endpoints.
- Developed Solidity smart contracts for Ethereum-based decentralized applications and performed source code security analysis of Solidity smart contracts and Node.js chaincode to identify vulnerabilities, logic flaws, and compliance gaps.

C-DAC: Centre for Development of Advanced Computing

Project Engineer | Pune, Maharashtra, India | Oct 2021 - Jun 2026

- Led the MeitY-funded initiative to build the National Blockchain Framework (NBF) for India's National Blockchain Service - full technical build detailed under Projects.

- Co-led a 7-person team building a full-stack Blockchain Security Assessment Portal (React, Node.js, Express, MongoDB) automating evaluations across 10-15 APIs, apps, containers, and databases.

Huptech Web Private Limited

Jr. UI Designer and Developer | Ahmedabad, Gujarat, India | Jun 2019 - Sep 2019

- Built responsive front-end interfaces using HTML, CSS, JavaScript, jQuery, and Bootstrap.

PROJECTS

National Blockchain Framework (NBF) - National Blockchain Service Initiative, MeitY

C-DAC: Centre for Development of Advanced Computing | Oct 2021 - Jun 2026

- Designed and architected a unified, multi-organization Hyperledger Fabric framework (3 peer nodes, 2 organizations, 1 channel) across 2-3 test environments.
- Built REST APIs for a security audit integrating blockchain services.
- Designed and built NBFDBAudit, a Python-based scanner that detects 18+ vulnerability classes across 4-5 MongoDB/CouchDB environments (including 1 production).
- Led VAPT across 4-5 blockchain applications and 30-45 APIs, surfacing ~25 vulnerabilities per application on average; partnered with engineering on remediation.
- Authored and presented 5 peer-reviewed papers at IEEE ICBDS and I2CT conferences on blockchain and smart contract security.

Development and Deployment of E-waste Management Modules Using Blockchain Technology

Graduate School of Engineering and Technology, GTU (M.E. Dissertation) | Jun 2020 - Jul 2021

- Designed and built a decentralized Ethereum DApp using Solidity, Truffle, Ganache, and IPFS. Implemented secure, tamper-resistant ownership tracking across the e-waste lifecycle.
- Integrated a React.js frontend with MetaMask wallet connectivity for on-chain interactions.
- Conducted threat modelling and optimized the decentralized architecture for performance.

PUBLICATIONS

A Threat Modeling Approach for Securing Blockchain Framework - 2025 4th IEEE ICBDS

Security Assessment Framework and Evaluation for Blockchain Applications (SAFE-Block) - 2024 3rd IEEE ICBDS

Efficient Solution for NoSQL Database Security in Blockchain-Based Applications - 2023 2nd IEEE ICBDS

Smart Contract Vulnerability Detection Techniques for Hyperledger Fabric - 2023 8th IEEE I2CT

Smart Contract - Security Assessment Integrated Framework (SC-SIF) for Hyperledger Fabric - 2022 7th IEEE I2CT

EDUCATION

Master's Degree in Cybersecurity

Graduate School of Engineering and Technology, GTU, Ahmedabad, Gujarat, India | Sep 2019 - Jul 2021

Bachelor's Degree in Computer Engineering

Marwadi Education Foundation, Rajkot, Gujarat, India | Aug 2015 - May 2019

CERTIFICATION & ACHIEVEMENTS

- Certified Blockchain Security Professional - Blockchain Council, USA
- Recognized as Performer of the Month - C-DAC, Pune, Maharashtra, India
- Blockchain Technology for Government Officials - Future Skills PRIME, NIELIT, Kolkata, India